



Web-based banking system

Project Group - 2026-Y2-S1-MLB-B5G1-02	
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01. Introduction

01.01 Project Overview

Serendib Smart Bank is a secure web-based management system developed to provide banking services through an online platform. The system enables customers to perform essential banking activities such as account management, balance inquiry, fund transfers, beneficiary management, loan applications, and transaction tracking.

The system also provides dedicated portals for bank employees, managers, and administrators, allowing efficient customer management, loan processing, reporting, and system administration. The platform aims to improve accessibility, security, and operational efficiency for both customers and banking staff.

Problem Looked After:

Traditional banking systems often require customers to visit physical branches for routine services such as fund transfers, loan applications, bill payments, and account inquiries. These processes can be time-consuming, inconvenient, and heavily dependent on manual operations.

With increasing demand for digital banking services, there is a need for a secure web-based banking platform that provides 24/7 access to banking services while ensuring data security, transaction integrity, and operational efficiency.

The proposed Serendib Smart Bank system addresses these challenges by providing a centralized online banking solution with secure authentication, OTP verification, AI-assisted support, and role-based management capabilities.

01.02 Project Objectives

The objectives of the project are:

- Develop a secure online banking platform.
- Provide role-based access control.
- Enable secure money transfers with OTP verification.
- Support online loan applications and approval workflows.
- Maintain accurate customer and transaction records.
- Provide administrative and management functionalities.
- Integrate an AI-powered customer assistant.

01.03 Target Users

- Customers
- Bank Employees
- Branch Managers
- System Administrators

01.04 Scope

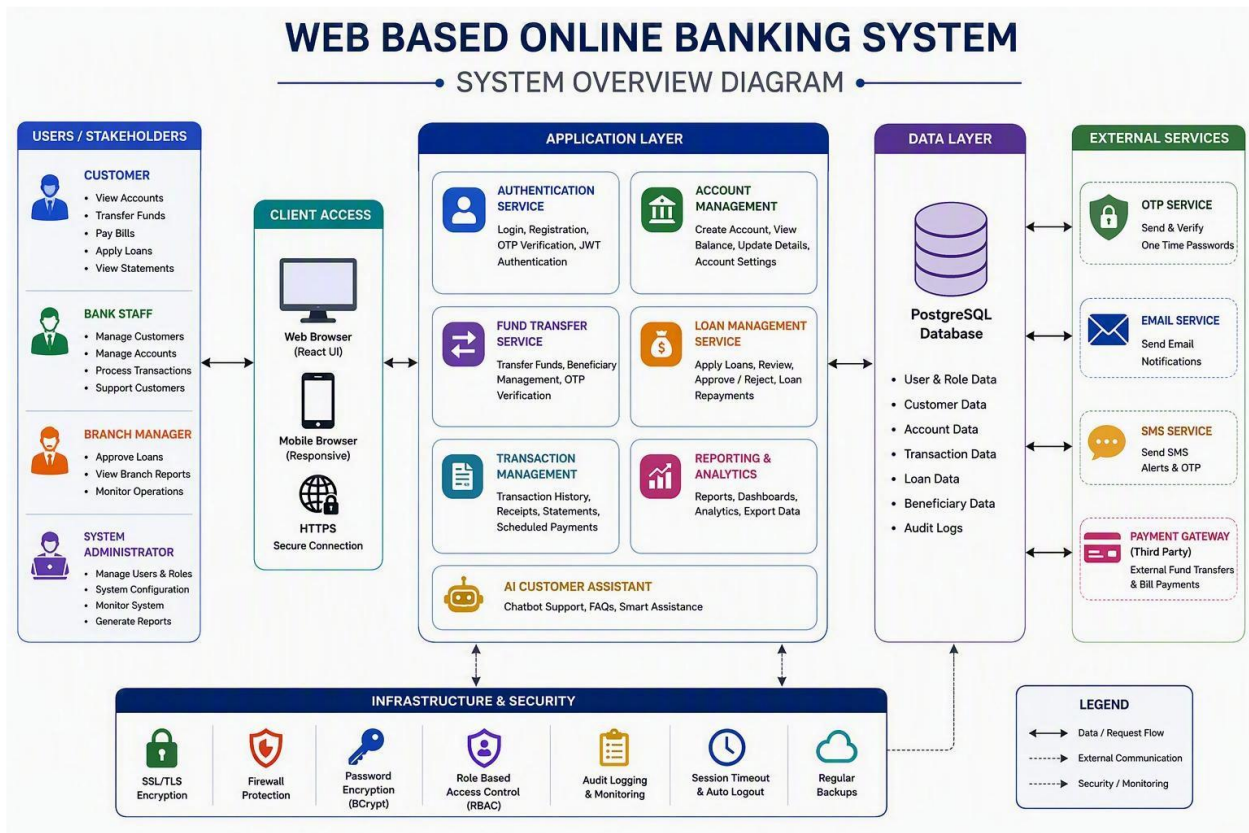
Included	Excluded
Customer management	Real ATM network integration
Account management	Real interbank network integration
Fund transfer & OTP verification	Real finance transactions
Beneficiary management	Mobile application
Loan management	
Employee management	
Reports and analytics	
AI customer assistant	
ATM integration simulation	
Interbank transaction simulation	
Bill Payment	
Card management	

01.05. Assumptions

The project assumes:

- Customers have internet access.
- Users possess basic computer literacy.
- OTP services can be simulated.
- PostgreSQL database will be available.
- Development will occur using Java Spring Boot and React.
- Real banking APIs are not required

02. System Overview Diagram



03. Functional Requirements

FR – 01. User authentication

- Login and logout functionality

FR – 02. Account management

- View account details and balances

FR – 03. Fund transfer

- Transfer funds securely using OTP

FR – 04. Beneficiary management

- Manage saved recipients

FR – 05. Loan management

- Apply, approve and track loans

FR – 06. User & role management

- Manage users and permissions

FR – 07. Bill payment

- Pay utility and service bills

FR – 08. Card management

- Manage debit/credit cards

FR – 09. Feedback management

- Submit and manage feedback

FR – 10. Reporting

- Generate operational reports

FR – 11. ATM simulation

- Withdraw, deposit, balance inquiry and mini statement

FR – 12. Interbank transfer simulation

- Transfer funds to external bank accounts

FR – 13. AI banking assistant

- Provide intelligent support

04. Non-Functional Requirements

Security

- Password shall be encrypted using BCrypt
- OTP verification shall be required for transactions
- Sessions shall automatically expire after 10 minutes of inactivity

Performance

- System response time shall be under 3 seconds
- Transactions shall be processed efficiently

Reliability

- Database backups shall be maintained
- Failed transactions must be recoverable

Availability

- System should operate 24/7

Usability

- User interfaces shall be simple and intuitive
- Navigation shall be user-friendly

Scalability

- Database should support future growth

05. Major Stakeholders / Users

Customer

Use online banking services

- Responsibilities
 - View accounts
 - Transfer money
 - Apply for loans
 - Pay bills
 - Manage cards
 - Submit feedback
 - Use AI assistance

Bank employee

Manage customer-related operations

- Responsibilities
 - Customer verification
 - Loan processing
 - Customer support
 - Manage customer inquiries
 - Review loan and transactions

Branch manager

Supervises branch operations

- Responsibilities
 - Approve high value loans
 - View branch reports
 - Employee supervision

System administrator

Maintains the entire system

- Responsibilities
 - User management
 - Security monitoring
 - Report generation

06. Major Functions

- **User management & bill payment**
 - Navishika D. M. N. N.
 - Customer user management
 - Utility bill payment
- **Account management**
 - Wanniararchchi U.P.
 - Account viewing
 - Balance Inquiry
- **Fund transactions and OTP verification**
 - Wijerathna T. L. R. B.
 - Fund transfer
 - OTP verification
 - Transaction verification
- **Loan management**
 - Jayasinghe W. M. S. B.
 - Apply loans
 - Review loans
 - Approve loans
- **Transaction management & beneficiaries**
 - Sanketh W. A. N.
 - Transaction history
 - Scheduled transactions
 - Beneficiary management
- **Card, feedback and report management**
 - Vidara M.K.A.O.
 - Card services
 - Customer feedback
 - Reporting

07. Minor Functions

- Login
- Logout
- Password reset
- Profile update
- Notification management
- Audit logging
- Search functionality

08. System Constraints

Technical constraints

- Developed using Java
- Web-based application only
- PostgreSQL database

Time constraints

- Must be completed within semester schedule

Resource constraints

- Limited access to real banking APIs

Security constraints

- No real financial transactions
- Simulated OTP service

09. Security Features

- **BCrypt Password Encryption**

Passwords stored as hashes.

- **OTP Verification**

Required for transactions.

- **Role-Based Access Control**

Customer, Employee, Manager, Admin.

- **Session Timeout**

10-minute inactivity timeout.

- **Audit Logging**

Tracking critical system activities.

- **Secure Database Access**

Prevent unauthorized access.

10. Timeline

Week	Activity
03	Proposal submission
04	Requirements analysis
05	UML diagram
06	Database design
07	Authentication module
08	Customer module
09	Transaction module
10	Progress evaluation
11	Loan Module
12	AI assistant
13	Testing
14	Final <u>submission</u>

11. Ethical Considerations

- **Customer Privacy**

Protect customer information.

- **Secure Data Usage**

Prevent misuse of personal data.

- **AI Transparency**

Users should know when interacting with AI.

- **Fair Access**

Provide equal access to services.

- **Responsible Development**

Follow secure software engineering practices.

12. Conclusion

The Serendib Smart Bank Web-Based Banking Management System aims to provide a secure, efficient, and user-friendly banking platform for customers and staff. The project incorporates modern software engineering principles including role-based access control, secure transaction processing, Agile development practices, database management, design patterns, and ethical considerations. The proposed system will demonstrate the practical application of software engineering methodologies while delivering a realistic banking solution.